13 Given that n can be any integer such that $n > 1$, prove that $n^2 - n$ is never an odd number.

$n^2 - n = n(n - 1)$

1 mark
Factorise as n(n − 1).

n and n − 1 are consecutive integers.

One is even, so their product is even.

1 mark
Explain why the product is always even.

Therefore $n^2 - n$ cannot be odd.

(Total for Question 13 is 2 marks)